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SUBJECT: REVISED CURRICULUM FOR DEGREE PROGRAMS IN CYBER SECURITY

The Higher Education Commission (HEC) of Pakistan, as mandated by its law, provides guidance to Higher Education Institutions (HEIs) on curricula for tertiary education levels in alignment with the National Qualifications Framework (NQF). To address evolving academic trends and market demands, HEC has revised the curriculum standards for Cyber Security degree programs at NQF levels 5, 6 and 7. These updated standards are aligned with HEC's Undergraduate Education Policy V 1.1 (2023), ensuring coherence with national priorities and adherence to international benchmarks.

2. The revised curricula for Cyber Security degree programs (attached), incorporating the option of specializations with advanced electives, are hereby notified. Universities offering this program are advised to align their Cyber Security curricula with these updated standards at the earliest. The finalized course contents should be developed according to the provided curriculum framework and be submitted electronically to this office. An electronic copy of this document can be found on HEC's official website.

3. Through the effective implementation of these updated standards, HEC aims to equip Cyber Security graduates with a comprehensive understanding of core principles such as network security, ethical hacking, digital forensics, cryptography, cyber threat intelligence, and secure software development. By fostering innovation, analytical thinking, and professional responsibility, these graduates will contribute to strengthening national security, safeguarding digital assets, and supporting sustainable development within the rapidly evolving technological landscape. Ultimately, these efforts support the goal of creating a resilient, secure, and digitally empowered society both locally and globally.

HAKIM ALI TALPUR

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CURRICULUM
FOR
CYBER SECURITY PROGRAMS
ASSOCIATE DEGREE
BS & MS
(2026)



HIGHER EDUCATION COMMISSION
ISLAMABAD – PAKISTAN

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PREFACE

The revised Cyber Security Curriculum (2026) has been developed in accordance with the Higher Education Commission's Undergraduate Education Policy (UEP) Version 1.1, with the aim of aligning academic standards with international best practices in the rapidly evolving field of cybersecurity. This comprehensive four-year undergraduate program is designed to equip students with a robust foundation in core cybersecurity principles, complemented by specialization areas such as network security, ethical hacking, digital forensics, cryptography, and cloud security.

The curriculum emphasizes a balanced integration of theoretical knowledge and practical skills. Students engage in hands-on projects, internships, and research activities across eight semesters, totaling a minimum of 133 credit hours, with flexibility up to 148 credit hours. The program includes mandatory general education courses to foster holistic development, along with discipline-specific core courses that cover essential areas such as programming, data structures, computer networks, operating systems, and cybersecurity-specific domains.

Specialization electives enable students to tailor their learning pathways according to their career aspirations, whether in threat intelligence, malware analysis, or cloud security, among others. Additionally, the curriculum promotes industry-recognized certifications, interdisciplinary courses, and field experience, ensuring graduates are well-prepared to meet the challenges of the digital age. This program aims to produce responsible, innovative, and adaptable cybersecurity professionals capable of addressing emerging threats and safeguarding digital assets across sectors such as finance, healthcare, government, and defense. Emphasizing ethical principles, legal obligations, and societal responsibilities, it also prepares students to become leaders and lifelong learners in the cybersecurity domain.

The development of this curriculum was driven by the collaborative efforts of the National Curriculum Revision Committee (NCRC), comprising academia, industry experts, and cybersecurity researchers. Their insights and expertise have ensured that this curriculum remains relevant, comprehensive, and aligned with national and international standards.

The Higher Education Commission recognizes the dedication of all contributors to this curriculum, which aims to foster academic excellence, research, and professional growth in cybersecurity, ultimately supporting Pakistan's digital security and technological advancement.

Dr. Amjad Hussain
Director General
Academics Division

GUIDING PRINCIPLES

MINIMUM STANDARDS

The curriculum standards and guidelines prescribed under this policy are mandatory at minimum level. Universities or the concerned departments may however set higher standards provided that the standards prescribed herein are not reduced or compromised.

COURSE SEQUENCE, TITLES AND CREDITS

For BS Cyber Security, the sequence of courses prescribed under this document is logically arranged and is suggestive only. The offering department may rearrange the sequence and alter the course titles and credits provided that the essence of the courses prescribed in policy remains intact. The department may add more courses as and when required subject to approval of university's relevant statutory body.

COURSE LEARNING OUTCOMES

Course learning outcomes (CLOs) are the bare minimum standards of learning that students must achieve upon completing a specific course. These outcomes serve as essential benchmarks, ensuring consistency in the quality of education across institutions. The CLOs prescribed herein represent the minimum level of competency and understanding expected from students. While these standards must not be compromised, departments are encouraged to enhance the rigor of the CLOs by incorporating additional learning outcomes, provided these do not alter the essence of the prescribed standards. In this policy, CLOs are exclusively developed for major field courses within the program. For interdisciplinary courses, departments offering these courses are responsible for developing their CLOs in alignment with their respective disciplines and program requirements. For general education courses as required under the HEC Undergraduate Education Policy V 1.1., departments may adopt the CLOs prescribed in the HEC-developed model courses.

COURSE SYLLABUS

This document serves as a comprehensive guideline describing the course learning outcomes (CLOs) for each course offered in the Associate Degree and BS in Cyber Security as minimum standards. The offering department is mandated to meticulously prepare, modify, and tailor the syllabus of each course, ensuring alignment with the stipulated learning outcomes. It is in this regard imperative that the department utilizes instructional, reference, and reading materials that it deems appropriate to effectively meet the CLOs.

GENERAL EDUCATION

For Associate degree and BS in Cyber Security, the courses prescribed for General Education component must mandatorily be offered with the same titles and credits as prescribed under HEC Undergraduate Education Policy V 1.1. The concerned departments may adopt and follow the learning outcomes and study contents developed by HEC for these courses as available on its website.

DISTRIBUTION OF COURSES IN SEMESTERS

Universities can redistribute courses across different semesters. This adjustment can be made based on factors such as teaching staff availability and other facilities. The course distribution outlined is not rigid and is subject to modification by the concerned universities.

IMPLEMENTATION

HEIs are advised to implement this revised scheme of studies and align their degree programs and degree nomenclature accordingly, with effective from Fall 2026.

REQUIREMENT OF FIELD EXPERIENCE / INTERNSHIP

It is a mandatory degree award requirement of three (03) credit hours for BS Cyber Security. Internship of six (06) to eight (08) weeks (preferably undertaken during semester or summer break) must be graded by a faculty member in collaboration with the supervisor in the field. This requirement cannot be substituted with additional course work, capstone or project work. Details are given at Annex-A.

REQUIREMENT OF CAPSTONE PROJECT

It is a mandatory degree award requirement of six (06) credit hours for BS Cyber Security. A capstone project is multifaceted body of work that serves as a culminating academic and intellectual experience for students. The capstone project must be supervised and graded by a faculty member as per the protocols prescribed by the concerned department. This requirement cannot be substituted with additional course work or internship.

REQUIREMENT OF CERTIFICATIONS

It is a mandatory requirement of three (3) credit hours for BS Cyber Security students to earn industry-relevant certifications, preferably through recognized national or international certification bodies, as outlined in **Annex-B**. These certifications are aimed at enhancing employability by equipping graduates with practical, job-ready skills in line with evolving market demands. The certifications may be considered for credit substitution (minimum of one and maximum of six credit hours) upon course mapping and approval by the relevant statutory bodies of the university. In BS Cyber Security program, students may opt to replace certain electives with internationally recognized certifications equivalent to three (03) credit hours. This requirement cannot be substituted with additional course work, internship, or project work.

ELECTIVES

- a) In accordance with the National Qualifications Framework, the department is required to offer at least 6 electives comprised of 18 credit hours i.e., approx.. 25% of the major i.e., BS Cyber Security comprised of 78 credit hours, to meet the criteria of nomenclature with specialization. Where the department increases the range of major beyond 78, the number of electives will accordingly be adjusted.
- b) Where the electives are opted from the general pool, the degree will be awarded as BS Cyber Security in its generic form and without any specialization. Example: Bachelor of Science in Cyber Security.
- c) Where the electives are opted from within a single specialization domain, the degree will be offered as Bachelor of Science in Cyber Security (with name of specialization) in accordance with the National Qualifications Framework (2015). Example: Bachelor of Science in Cyber Security (Network and Infrastructure Security).

ENTRY PROVISION AT UNDERGRADUATE LEVEL

PATHWAY FOR GRADUATES WITH ASSOCIATE DEGREE

- a) Only the students having 14 years qualification in Computing are allowed admission in the fifth semester of the BS in Cyber Security. However, as per revised scheme such students are also required to complete deficiency courses from 15-18 credits through bridging semester prior to enrollment in 5th Semester of BS Cyber Security Program. The deficiency is for the purpose to make prior learning before 5th semester at par with scheme of BS Cyber Security program.
- b) The minimum eligibility for admission in the fifth semester through above-mentioned pathway is 2.00/4.00 CGPA in the prior qualification i.e., Associate Degree in Cyber

Security/Computing. The concerned university may, however, set higher eligibility and admission criteria for admission in the fifth semester of BS in Cyber Security.

BACHELOR OF SCIENCE IN CYBER SECURITY

PROGRAM DESCRIPTION

The BS in Cyber Security is a comprehensive four-year undergraduate degree program designed to align with the HEC Undergraduate Education Policy V 1.1. This program aims to equip students with a solid foundation in core cybersecurity principles, along with specialized knowledge in emerging areas such as network security, ethical hacking, digital forensics, cryptography, cyber threat intelligence, embedded systems security, and software security.

The curriculum emphasizes both theoretical understanding and practical skills, fostering innovative problem-solving, risk assessment, and the development of secure technological solutions. Students will engage in hands-on projects, internships, and research activities to get prepared for real-world challenges faced by organizations in sectors such as finance, healthcare, government, and defense.

The program fosters ethical and professional responsibility, teamwork, and effective communication, preparing graduates to be adaptable, responsible cybersecurity professionals and leaders. Graduates will be equipped to address evolving cyber threats and safeguard digital assets in a rapidly changing technological landscape.

STANDARD NOMENCLATURE

For the sake of standardization, the standard nomenclature of four-year undergraduate degree programs in Cyber Security must be **“Bachelor of Science in Cyber Security” with its short form as “BS Cyber Security”**.

Note: The scheme of study prescribed for the four-year undergraduate degree in Cyber Security is based on a total of six (06) electives. Where these courses are opted from the general pool of electives, the degree will be titled “Bachelor of Science in Cyber Security” with its short form as “BS Cyber Security” in its generic form and without any specialization. Whereas, if all the electives are opted from a single specialization domain, the degree will be titled Bachelor of Science in Cyber Security with the name of specialization in parenthesis in accordance with the National Qualifications Framework (2015). Further specializations as per given framework may be added by HEIs based on availability of faculty, infrastructure and resources. The area of specializations accordingly shall be reflected clearly on the transcript.

PROGRAM LEARNING OUTCOMES /GRADUATE ATTRIBUTES

Cyber Security programs prepare students to attain educational objectives by ensuring that students demonstrate achievement of the following outcomes:

1. **Academic Education**
Prepare graduates as cyber security professionals with a strong foundation in computing and cyber security.
2. **Knowledge for Solving Cyber Security Problems**
Apply knowledge of mathematics, computing, and cyber security to analyze and solve cyber security problems.
3. **Problem Analysis**
Identify, formulate, investigate relevant literature, and analyze complex cyber security problems, reaching well-justified conclusions using principles of mathematics, computing, and cyber security.

4. **Secure Solutions Development**
Design and develop solutions that meet specified requirements with appropriate consideration of security, legal, societal, environmental, and safety constraints.
5. **Modern Tool Usage**
Select, adapt, and apply appropriate cyber security techniques, methodologies, and modern tools for threat analysis, modeling, monitoring, and mitigation of cyber security problems.
6. **Individual and Team Work**
Work effectively as an individual and as a member or leader of diverse and multidisciplinary teams in cyber security and related professional settings.
7. **Communication**
Communicate effectively on cyber security issues, risks, and solutions with technical professionals, stakeholders, and society at large using appropriate communication methods.
8. **Professionalism and Society**
Understand and evaluate societal, health, safety, and cultural issues related to cyber security, and recognize the responsibilities of cyber security professionals in local and global contexts.
9. **Legal and Ethical Understanding**
Demonstrate understanding of ethical principles, professional responsibilities, and legal obligations relevant to cyber security practice.
10. **Lifelong Learning**
Recognize the importance of continuous, self-directed learning in the rapidly evolving field of cybersecurity and demonstrate the ability to pursue ongoing professional development to remain current and effective.

ELIGIBILITY CRITERIA

Higher Secondary School Certificate/A-levels (involving minimum 12 years of schooling) or an IBCC equivalent qualification with at least 50% marks and having studied Mathematics is the basic eligibility requirement for admission in BS Cyber Security Program.

- The students who have not studied Mathematics at intermediate level have to pass deficiency courses of Mathematics (06 credits) in first year including zero semester.

The admitting university may set minimum eligibility scores and may conduct entry / admission test through its own testing body or an external testing services provider of repute as per the screening, admission and merit calculation criteria approved by its statutory bodies.

PROGRAM STRUCTURE OF BS CYBER SECURITY

The BS Cyber Security is structured in accordance with the provisions of the HEC Undergraduate Education Policy V 1.1. and comprises of minimum 133 credit hours spread over eight (08) regular semesters. Universities may offer courses up-to maximum of 148 credit hours provided that the total number of credit hours are reasonably set to achieve the Program Learning Objectives (PLOs) without putting undue burden on students.

Minimum Credit Hours		133
General Education Courses		34 credit hours (14 courses)
Discipline Related Courses / Major	Total	78 credit hours (23 courses)
	Computing Core Courses	30 credit hours (8 courses) Mandatory courses as reflected in the scheme of study, irrespective of the area of specialization (where applicable).

	Cyber Security Core Courses	27 credit hours (9 courses) Mandatory courses as reflected in the scheme of study, irrespective of the area of specialization (where applicable).
	Specialization Electives	18 credit hours (6 courses) (to be opted from either general pool of electives or from a particular specialization)
	Professional Certification	03 credit hours
Interdisciplinary/ Allied Courses		12 credit hours (4 courses)
Field Experience/Internship		03 credit hours
Capstone Project / FYP		06 credit hours
Program Duration		Minimum: 4 Years Maximum: 6 Years (Further extendable to another year subject to the approval of the university's statutory body following the provisions of HEC Undergraduate Education Policy 2023 V 1.1)
Semester Duration		16-18 weeks for regular semesters (1-2 weeks for examination) 8-9 weeks for summer semesters (1 week for examination)
Course Load (per semester)		16-21 credit hours for regular semesters (Excluding the last or second last semester) Credit hours to be offered in Summer/winter semesters, as per HEC guidelines (For remedial/deficiency/failure/repetition courses only)
3 Credit Hours (Theory)		3 classes (1 hour each) OR 2 classes (1.5 hours each) OR 1 class (3 hours) per week throughout the semester.
1 Credit Hours (Practical Work)*		1 (one) Credit hour of practical work requires 3 (three) contact hours per week throughout the semester.
Policy for Probation in Semester		i) The students acquiring less than 2.00/4.00 GPA in a semester but passing in all papers will be promoted with the condition to achieve more than 2.0 GPA in the next

	semester and s/he will be put on probation for the next semester.
	ii) The students acquiring GPA 1.7 and above but failing in any paper(s) will be placed on probation and promoted to the next semester conditionally. They will have to be registered for summer semester to improve the grade.
	iii) Students acquiring GPA less than 1.7 in two consecutive semesters and failing in any paper(s) even after attending summer semester for one academic year will be dropped from university rolls.

GENERAL EDUCATION COURSES: 34 CREDITS (14 COURSES)

As per HEC UGE Policy V 1.1 and subsequent notifications, following courses are mandatory to be part of every undergraduate degree program including Associate Degree, hence the same are included in the schemes of BS Cyber Security;

Serial No.	GE Course Category	No. of Courses	Credit Hours
1	Arts & Humanities *	1	02
2	Natural Sciences *	1	03 (2+1)
3	Social Sciences *	1	02
4	Functional English, Expository Writing	2	06
5	Quantitative Reasoning	2	06
6	Islamic Studies / Ethics	1	02
7	Ideology & Constitution of Pakistan	1	02
8	Application of ICT	1	03 (2+1)
9	Entrepreneurship	1	02
10	Civics and Community Engagement	1	02
11	Pakistan Studies	1	02
12	Fehm e Quran	1	02
	Total	14	34

Note: In the course categories from Serial Number 4-12, developed by Academics Division, HEC through Course Design Committees have been uploaded at HEC website and can be accessed by the following weblink:

<https://www.hec.gov.pk/english/services/universities/cr/Pages/Framework-for-Revision-of-Curricula.aspx>

In the course categories from Serial Number 1-3, the HEIs may choose a course from the pool of subjects under the category approved by statutory bodies of the university.

MAJOR COURSES (COMPULSORY): 78 CREDIT HOURS (23 COURSES)

The Core Mandatory courses in a BS Cyber Security program help to provide students with a strong and comprehensive foundation in the fundamental areas of BS Cyber Security program.

COMPUTING: 30 CREDIT HOURS (8 COURSES)

S.No.	Courses	Credit Hours
1.	Programming Fundamentals	3+1
2.	Object Oriented Programming	3+1
3.	Data Structures and Algorithms	3+1
4.	Database Systems	3+1
5.	Digital Logic Design	3+1
6.	Operating Systems	3+1
7.	Computer Networks	2+1
8.	Computer Organization and Architecture	2+1
	Total	30 credit hours

CYBER SECURITY: 27 CREDIT HOURS (9 COURSES)

S.No.	Courses	Credit Hours
1.	Introduction to Cyber Security	2+1
2.	Network Security	2+1
3.	Information Security Management	3+0
4.	Secure Software Design and Development	2+1
5.	Digital Forensics and Incident Response	2+1
6.	Cryptography	2+1/3+0
7.	Vulnerability Assessment and Penetration Testing	2+1
8.	Malware Analysis and Reverse Engineering	2+1
9.	Cloud and Application Security	2+1
	Total	27 credit hours

RECOMMEDED LIST OF INTERDISCIPLINARY COURSES:

As per HEC UGE Policy V 1.1, interdisciplinary courses of Min 12 credit hours are required in four year BS Cyber Security program, to complement holistic understanding of the major. The item was thoroughly deliberated in the NCRC of Cyber Security it was emphasized that in order to effectively utilize the compartment of interdisciplinary courses, the list be given so that no important area is left unattended. Universities are at liberty to add more courses to it after approval of HEIs' relevant statutory bodies. However, the first course in the below mentioned list is mandatory as decided by the NCRC.

S. No	Interdisciplinary / Allied Courses*
1.	Artificial Intelligence (Mandatory)
2.	Number Theory and Finite Fields
3.	Environmental Science
4.	Discrete Structures
5.	Introduction to Game Development
6.	Linear Algebra and Differential Equations
7.	Interpersonal Skills
8.	Media and Information Literacy
9.	Probability and Statistics
10.	Organizational Behavior
11.	Project Management
12.	Calculus and Analytical Geometry
13.	Introduction to Psychology
14.	Digital Marketing and E-Commerce
15.	Digital Image Processing
16.	Financial Technology (FinTech) Fundamentals
17.	Space Technology and Applications
18.	Internet Economics
19.	Smart Cities and Urban Informatics
20.	Technology and Sustainable Development
21.	Digital Media and Communication Studies
22.	International Relations

SCHEME OF STUDIES

The suggestive scheme of studies for BS Cyber Security program is given below:

SEMESTER-I			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Introduction to Cyber Security	3 (2+1)	Major
2	Programming Fundamentals	4 (3+1)	Major
3	Quantitative Reasoning-I*	3 (3+0)	General Education
4	Functional English*	3 (3+0)	General Education
5	Applications of Information and Communication Technologies*	3 (2+1)	General Education
6	Islamic Studies (Religious Education / Ethics for non-Muslim students) *	2 (2+0)	General Education
Total Credits (18)			

SEMESTER-II			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Object Oriented Programming	4 (3+1)	Major
2	Natural Science****	3 (2+1)	General Education
3	Digital Logic Design	4 (3+1)	Major
4	Quantitative Reasoning-II*	3 (3+0)	General Education
5	Arts and Humanities**	2 (2+0)	General Education
6	Ideology and Constitution of Pakistan*	2 (2+0)	General Education
Total Credits (18)			

SEMESTER-III			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Expository Writing*	3 (3+0)	General Education
2	Secure Software Design and Development	3 (2+1)	Major
3	Data Structures and Algorithms	4 (3+1)	Major
4	Computer Networks	3 (2+1)	Major
5	Operating Systems	4 (3+1)	Major
6	Fehm-e-Quran – I (for Muslim Students)*	1 (0+1)	General Education
7	Civics and Community Engagement*	2 (2+0)	General Education
Total Credits (20)			

SEMESTER-IV			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Computer Organization and Architecture	3 (2+1)	Major
2	Network Security	3 (2+1)	Major
3	Cryptography	3 (2+1/3+0)	Major
4	Database Systems	4 (3+1)	Major
5	Vulnerability Assessment and Penetration Testing	3 (2+1)	Major
6	Entrepreneurship*	2 (2+0)	General Education
7	Fehm-e-Quran – II (for Muslim Students)*	1 (0+1)	General Education
Total Credits (19)			

SEMESTER-V			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Artificial Intelligence IDS – I*****	3 (2+1)	IDS - Mandatory
2	Digital Forensics and Incident Response	3 (2+1)	Major
3	Malware Analysis and Reverse Engineering	3 (2+1)	Major
4	Social Science***	2 (2+0)	General Education
5	Information Security Management	3 (3+0)	Major
6	Elective-I*****	3	Major (Specialization)
7	Pakistan Studies*	2 (2+0)	General Education
Total Credits (19)			

SEMESTER-VI			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Cloud and Application Security	3 (2+1)	Major
2	IDS – II*****	3 (3+0)	IDS
3	Elective-II*****	3	Major (Specialization)
4	Elective-III*****	3	Major (Specialization)
5	Final Year Project I (Proposal Defense) (As per protocols given) *****	1	FYP/Capstone Project
6	Elective-IV*****	3	Major (Specialization)
Total Credits (16)			

SEMESTER-VII			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	IDS – III*****	3 (3+0)	IDS
2	IDS – IV*****	3 (3+0)	IDS
	Elective-V*****	3	Major (Specialization)
3	Elective-VI*****	3	Major (Specialization)
4	Final Year Project II (Design and Development) (As per protocols given) +++	2	FYP/Capstone Project
5	Professional Certification++	3	Certification
Total Credits (17)			

SEMESTER-VIII			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Final Year Project-III (Demonstration and Presentation) +++ (As per protocols given)	3	FYP/Capstone Project
2	Field Experience	3	Field Experience
Total Credits (6)			

- * HEC designed model courses for general education may be used by the university.
- ** HEC designed model course may be used by the university, OR the university / offering department may offer any course in the broader category of **“Arts & Humanities”** including but not limited to a course of regional or international language such as Chinese, Arabic, French, Spanish etc. or any other course such as Philosophy, History etc.
- *** HEC designed model course may be used by the university, OR the university / offering department may offer any course in the broader category of **“Social Sciences”** including but not limited to a course of Psychology, Sociology, Anthropology etc.
- **** HEC designed model course may be used by the university, OR the university / offering department may offer any course in the broader category of **“Natural Sciences”** which should have relevance to the purpose of the degree program.
- ***** The university / concerned department must offer at least 4 interdisciplinary courses from the recommended list provided in this document, or any other such course to enhance the interdisciplinary understanding of the students, keeping in view the available academic, human and infrastructural resources of the department.
- ***** Read in conjunction with guidance given on “Standard Nomenclature” in this document, the university / concerned department may offer any 6 advanced courses from either the general pool or from any one of the specializations as electives keeping in view its available academic, human and infrastructural resources.

- ++ BS Cyber Security students are required to complete professional certification(s) (equivalent to 3 credit hours in total) over the period of four-year program as a mandatory condition for degree completion. A certification will be considered equivalent to 1 credit hour if it comprises at least 16 hours of study. The respective department will guide students in selecting relevant certifications, ensuring alignment with current market needs and the program's objectives. **Details given at Annex-B**
- +++ The Final Year Project may be split in 6th, 7th and 8th Semester. Likewise the Field Experience may be offered during summer/winter semester.

DEGREE AWARD REQUIREMENTS

The following minimum requirements are prescribed for the award of BS Cyber Security:

- All courses in the General Education category with titles and credit hours as prescribed in HEC Undergraduate Education Policy V 1.1. must be completed.
- Minimum of 133 credit hours (including internship / field experience, certifications and capstone project) as prescribed in this policy document must be completed.
- Capstone / research project of 6 credit hours must be completed in accordance with HEC Undergraduate Education Policy V 1.1. This requirement cannot be substituted with additional coursework or internship.
- Internship / Field Experience of 3 credit hours must be completed in accordance with HEC Undergraduate Education Policy V 1.1. This requirement cannot be substituted with additional coursework, capstone, research or project work.

SPECIALIZATIONS IN BS CYBER SECURITY PROGRAM:

After detailed consultation in the NCRC, following specializations under BS Cyber Security are recommended. HEIs may add more specializations as per availability of faculty, infrastructure and resources keeping in view the framework after approval of their statutory bodies;

- Network and Infrastructure Security
- Cryptography
- Digital Forensics and Incident Response
- IoT and Embedded Systems Security
- Governance, Risk and Compliance

GUIDELINES FOR SPECIALIZATIONS:

A total of 18 credit hours in the BS Cyber Security program are allocated to elective courses, which define the specializations offered under the program. Each specialization cluster provides a broad range of elective courses, from which students must select 6 courses to complete the required 18 credit hours.

The specialization title may appear either on the degree (e.g., BS Cyber Security – Network and Infrastructure Security) or only on the transcript, as per the approval of the university's statutory bodies.

Universities also have the flexibility to offer a generic BS Cyber Security program, wherein students may choose electives from any cluster without committing to a specific specialization. In such cases, no specialization title will appear on the degree or transcript.

1. Network and Infrastructure Security

SR NO	COURSES	Credit Hour
1.	Cloud Security	3

2.	Data Center Security	3
3.	Mobile and Wireless Network Security	3
4.	Network and System Administration	3
5.	Machine Learning for Network Security	3
6.	Virtualization Security	3
7.	Network Forensics	3
8.	Endpoint Security	3
9.	Network Security Monitoring	3
10.	Security Automation for Network Operations	3
11.	Security Operations Center (SOC) and Threat Monitoring	3

2. Cryptography

SR NO	COURSES	Credit Hour
1.	Information Privacy and Security	3
2.	Mathematical Foundations for Cryptography	3
3.	Symmetric Key Cryptography	3
4.	Asymmetric Key Cryptography	3
5.	Cryptographic Hash Functions and Blockchain Applications	3
6.	Cryptographic Protocols and Key Management	3
7.	Cryptanalysis and Attack Techniques	3
8.	Secure Systems and Applied Cryptography	3
9.	Advanced Cryptography	3
10.	Post-Quantum and Emerging Cryptography	3
11.	Data Security and Privacy Engineering	3

3. Digital Forensics and Incident Response

SR NO	COURSES	Credit Hour
1.	Computer Forensics	3
2.	Network Forensics	3
3.	Smart Devices and IoT Forensics	3
4.	Cloud Forensics and Log Analysis	3
5.	Cyber Threat Intelligence	3
6.	Open-Source Intelligence Techniques	3
7.	Incident Response Orchestration and Automation	3
8.	Multimedia Forensics	3

9.	eDiscovery Forensics and Moot Courts	3
10.	AI-Driven Digital Forensics	3

4. IoT and Embedded Systems Security

SR NO	COURSES	Credit Hour
1.	Introduction to Embedded System Security	3
2.	Internet of Things (IoT) Security	3
3.	Mobile Device Security	3
4.	Industrial IoT and Critical Infrastructure Security	3
5.	Smart and Autonomous Vehicle Security	3
6.	Avionics Security	3
7.	Smart Devices and IoT Forensics	3
8.	Hardware Design Security	3
9.	Embedded Operating System Security	3
10.	Wireless Networks Security	3
11.	Industrial Control Systems Security	3

5. Governance, Risk and Compliance

SR NO	COURSES	Credit Hour
1.	Cyber Governance	3
2.	Cyber Security Risk Management	3
3.	Cyber Warfare	3
4.	Cyber Laws, Regulations and Ethics	3
5.	Compliance Standards and Frameworks	3
6.	Cyber Security Audit and Assurance	3
7.	Financial Systems Security	3
8.	Privacy Engineering and Data Protection	3
9.	Third-Party and Supply Chain Risk Management	3
10.	Human Behaviours in Cyber Security	3
11.	Cyber Security Economics, Policy & Strategy	3

Note: It is recommended that the above mentioned specialization courses be offered with labs/practical work where required.

ASSOCIATE DEGREE CYBER SECURITY

STANDARD NOMENCLATURE

For the sake of standardization, it was decided by NCRC that the undergraduate degree program (NQF Level-05) shall have the title of **“Associate Degree Cyber Security”**.

ELIGIBILITY CRITERIA

Higher Secondary School Certificate/A-levels (involving minimum 12 years of schooling) or an IBCC equivalent qualification with at least 45% marks and having studied Mathematics is the basic eligibility requirement for admission in Associate Degree in Cyber Security.

- The students who have not studied Mathematics at intermediate level have to pass deficiency courses of Mathematics (06 credits) in first year including zero semester.

Additionally, candidates may be required to pass an entry test conducted by the university or admission authority, along with fulfilling any other criteria set by the institution, such as interviews or aptitude assessments.

PROGRAM STRUCTURE OF ASSOCIATE DEGREE CYBER SECURITY

The degree program follows HEC Undergraduate Education Policy 2023 V 1.1 and comprises a minimum of 04 regular semesters (02 years). Universities may offer courses consisting of a minimum of 72 credit hours provided that the total number of credit hours are reasonably set to achieve the Program Learning Outcomes.

Minimum Credit Hours		72
General Education Courses		34 credit hours (14 courses)
Discipline Related Courses / Major	Total	38 credit hours (11 courses)
	Core Courses	26 credit hours (8 courses)
	Specialization Electives	12 credit hours (4 courses)
Program Duration		Minimum: 2 Years Maximum: 3.5 Years
Semester Duration		16-18 weeks for regular semesters (1-2 weeks for examination) 8-9 weeks for summer semesters (1 week for examination)
Course Load (per semester)		15-21 credit hours for regular semesters Credit hours to be offered in Summer/winter semesters, as per HEC guidelines (For remedial/deficiency/failure/repetition courses only)

3 Credit Hours (Theory)	3 classes (1 hour each) OR 2 classes (1.5 hours each) OR 1 class (3 hours) per week throughout the semester.
1 Credit Hours (Practical Work)*	1 Credit hour of practical work requires three contact hours per week throughout the semester.

GENERAL EDUCATION COURSES: 34 CREDITS (14 COURSES)

As per HEC UGE Policy V 1.1 and subsequent notifications, following courses are mandatory to be part of every undergraduate degree program including Associate Degree, hence the same are included in the schemes of Associate degree Cyber Security;

S. No.	GE Course Category	No. of Courses	Credit Hours
1	Arts & Humanities *	1	02
2	Natural Sciences *	1	03 (2+1)
3	Social Sciences *	1	02
4	Functional English, Expository Writing	2	06
5	Quantitative Reasoning	2	06
6	Islamic Studies / Ethics	1	02
7	Ideology & Constitution of Pakistan	1	02
8	Application of ICT	1	03 (2+1)
9	Entrepreneurship	1	02
10	Civics and Community Engagement	1	02
11	Pakistan Studies	1	02
12	Fehm e Quran	1	02
	Total	14	34

MAJOR COURSES (COMPULSORY): 26 CREDIT HOURS (08 COURSES)

The Core Mandatory courses in Associate Degree Cyber Security program help to provide students with a strong and comprehensive foundation in the fundamental areas of computing and cyber security.

S. No.	Courses	Credit Hours
1	Introduction to Cyber Security	3 (2+1)
2	Programming Fundamentals	4 (3+1)
3	Data Structures and Algorithms	4 (3+1)
4	Computer Networks	3 (2+1)
5	Database Systems	3 (2+1)
6	Network Security	3 (2+1)
7	Operating Systems	3 (2+1)

8	Secure Software Design and Development	3 (2+1)
	Total	26 credit hours

POOL OF ELECTIVES: 12 CREDIT HOURS (04 COURSES)

Below is the recommended list of major elective courses for Associate degree Cyber Security. The department concerned may offer any four (04) courses from the following list or any other course relevant to the specialization of Cyber Security, as per the availability of academic resources:

S. No.	Courses	Credit Hours
1	Digital Forensics and Incident Response	3
2	Malware Analysis and Threat Intelligence	3
3	Cloud and Application Security	3
4	Artificial Intelligence for Cyber Security	3
5	Vulnerability Assessment and Penetration Testing	3
6	Open-Source Intelligence Techniques	3
7	Security Operations and Incident Response	3
8	Network Security Monitoring	3
9	Identity and Access Management	3
10	Information Security Management	3

Note: HEP's may add further courses in the above pool of electives after approval from their statutory bodies.

SCHEME OF STUDIES FOR ASSOCIATE DEGREE CYBER SECURITY

The standard scheme of studies for Associate degree Cyber Security is given below:

SEMESTER-I			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Introduction to Cyber Security	3 (2+1)	Major
2	Programming Fundamentals	4 (3+1)	Major
3	Quantitative Reasoning-I*	3 (3+0)	General Education
4	Functional English*	3 (3+0)	General Education
5	Applications of Information and Communication Technologies*	3 (2+1)	General Education
6	Islamic Studies (Religious Education / Ethics for non-Muslim students) *	2 (2+0)	General Education
Total Credits (18)			

SEMESTER-II			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Data Structures and Algorithms	4 (3+1)	Major
2	Natural Science**	3 (2+1)	General Education
3	Computer Networks	3 (2+1)	Major
4	Quantitative Reasoning-II*	3 (3+0)	General Education
5	Arts and Humanities**	2 (2+0)	General Education
6	Secure Software Design and Development	3 (2+1)	Major
Total Credits (18)			

SEMESTER-III			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Expository Writing*	3 (3+0)	General Education
2	Database Systems	3 (2+1)	Major
3	Operating Systems	3 (2+1)	Major
4	Networks Security	3 (2+1)	Major
5	Elective-I	3(2+1)	Elective
6	Fehm-e-Quran – I (for Muslim Students)	1 (0+1)	General Education
7	Social Science**	2 (2+0)	General Education
Total Credits (18)			

SEMESTER-IV			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Elective-II	3(2+1)	Elective
2	Elective-III	3(2+1)	Elective
3	Elective-IV	3(2+1)	Elective
4	Entrepreneurship*	2 (2+0)	General Education
5	Fehm-e-Quran – II (for Muslim Students)	1 (0+1)	General Education
6	Pakistan Studies*	2 (2+0)	General Education
7	Ideology and Constitution of Pakistan*	2 (2+0)	General Education
8	Civics and Community Engagement*	2 (2+0)	General Education
Total Credits (18)			

Note: Elective courses may be replaced with relevant recommended certifications listed in Annex B of this document. Universities shall determine the evaluation and acceptance criteria for such certifications through their statutory academic bodies, including the Board of Studies, Board of Faculty, and Academic Council.

MASTER OF SCIENCE IN CYBER SECURITY

PROGRAM DESCRIPTION

The Master of Science in Cyber Security is structured in accordance with the HEC Graduate Education Policy 2023 and is designed to provide students with advanced knowledge and practical skills in cybersecurity. The program builds a strong foundation in the principles, practices, and technologies necessary to protect information systems, networks, software, and digital assets from evolving cyber threats.

Through a combination of core studies, electives, and research-oriented activities, students gain specialized expertise in areas such as cryptography, cloud and distributed systems security, embedded and IoT systems, digital forensics, malware analysis, and incident response. The program emphasizes both theoretical understanding and analytical, problem-solving skills, preparing students to evaluate and implement effective security solutions in complex and dynamic environments.

Designed for students with a background in computer science, information technology, cyber security, or related disciplines, the program equips graduates to assess risks, design secure systems, implement security measures, and respond to cyber incidents across a variety of sectors. Additionally, it fosters the ability to analyze emerging threats, evaluate new security technologies, and develop innovative approaches to safeguard critical infrastructure.

Graduates are prepared for professional careers in the public sector, private industry, national security organizations, and research institutions, as well as for advanced academic research, ensuring readiness to meet the challenges of a rapidly evolving global cybersecurity landscape.

STANDARD NOMENCLATURE

To ensure uniformity, the standard nomenclature of all graduate degree programs (NQF level-7) qualifications in the field of Cyber Security will be either **"Master of Science in Cyber Security"** (with its short form as MS Cyber Security) subject to approval by relevant statutory bodies of the university.

PROGRAM LEARNING OUTCOMES

By the completion of a Master of Science (MS) in Cyber Security, the graduates will be able to:

- Synthesize and evaluate advanced cybersecurity knowledge, threat intelligence, and research to solve complex and evolving security challenges.
- Forecast, model, and analyze emerging cyber threats, vulnerabilities, and risks, and design resilient, scalable, and secure systems and policies.
- Select, integrate, and critically evaluate advanced cybersecurity tools and platforms to implement innovative and effective security solutions.
- Lead and collaborate in multidisciplinary teams to plan, execute, and drive complex cybersecurity initiatives, including incident response, security research, and policy development.
- Communicate cybersecurity research, strategies, and technical knowledge effectively to diverse audiences, supporting decision-making and policy implementation.

- Analyze complex ethical dilemmas and advance responsible cybersecurity innovation while upholding professional standards and considering societal, legal, and operational impacts.
- Continuously engage in professional development to remain at the forefront of emerging cybersecurity research, tools, and methodologies.

ELIGIBILITY CRITERIA

- An undergraduate degree involving 16 years of education in Cyber Security is the basic eligibility requirement for admission in the MS Cyber Security.
- Candidates having undergraduate degree involving 16 years of education in computing or relevant engineering discipline are also eligible for admission in MS Cyber Security subject to completion of deficiency courses up-to 12 credit hours (to be determined by the admitting university / department). Relevance of the prior qualification in this regard will also be determined by the admitting university / department.
- In addition to the basic eligibility, the admitting university is further required to conduct a rigorous admission test as an eligibility condition for admission to the program, with a passing score of 50% (OR) accept the GRE/HAT General/equivalent tests, with a passing score of 50%. The admitting university may also set minimum eligibility scores (above 50%) as per the screening, admission and merit calculation criteria approved by its statutory bodies.

PROGRAM STRUCTURE

The MS Cyber Security is structured in accordance with the provisions of the HEC Graduate Education Policy (GEP-2023). The standard structure of the program is as under:

Minimum Credit Hours	32
Course Work	Minimum 26 credit hours (10 courses) Core: 9 credit hours (3 courses) Electives: 15 credit hours (5 courses) General Education: 2 credit hours (2 courses)
Research Work / Thesis	Minimum 06 credit hours
Program Duration	Minimum: 1.5 Years (3 regular semesters) Maximum: 4 Years (8 regular semesters) Note: In case a student is unable to secure an MS within the prescribed timeframe and claims for extension in duration, the university may constitute appropriate authority and determine the causes of delay. In the event of force majeure (i.e., delay on account of circumstance beyond the control of student), the university may grant an extension in the period of award of MS degree in accordance with the duration limiting factor(s) and shall also take corrective measures in case the delay is caused by process or administrative reasons.
Semester Duration	16-18 weeks for regular semesters

	(1-2 weeks for examination) (1 week for examination)
Course Load (per semester)	10-12 credit hours for regular semesters
3 Credit Hours (Theory)	3 classes (1 hour each) OR 2 classes (1.5 hours each) OR 1 class (3 hours)

SCHEME OF STUDIES:

The standard scheme of studies for MS Cyber Security is given below:

SEMESTER I			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Data and Software Security*	3	Core
2	Advanced Networks Security*	3	Core
3	Research Methodology*	3	Core
4	Understanding of Holy Quran – I **	1 (0-1)	General Education
Total Credits (10)			

SEMESTER II			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Elective – I ***	3	Elective
2	Elective – II ***	3	Elective
3	Elective – III ***	3	Elective
3	Understanding of Holy Quran – II **	1 (0-1)	General Education
Total Credit Hours (10)			

SEMESTER III			
S. No	COURSE	CREDIT HOURS	CATEGORY
1	Elective – IV ***	3	Elective
2	Elective – V ***	3	Elective
3	Thesis ****/Course Work	6 ****	Research
Total Credit Hours (12)			

SEMESTER IV			
S. No	COURSE	CREDIT HOURS	CATEGORY
	Thesis ****/ Course Work	Continued	Research

- * These are the mandatory courses for the program, irrespective of the area of specialization.
- ** These are prescribed for all NQF level 5-8 qualifications as mandated vide HEC letter No.10-01/2022/Coord(Acad)/HEC/235 dated March 28, 2025.
- *** The university / concerned department may offer any advanced course in the field of Cyber Security as an elective, as per its available academic, human and infrastructural resources (a recommended list of electives is available below).
- **** Research work for thesis must be conducted by students individually in accordance with the university's policy as approved through its statutory bodies provided that the same is in accordance with the HEC Graduate Education Policy (2023).

LIST OF ELECTIVE COURSE:

Below is the recommended list of major elective courses for MS Cyber Security. The department concerned may offer any five (05) courses from the following list or any other course relevant to Cyber Security, as per the availability of academic resources:

1. Compliance and Auditing in Cyberspace
2. Security of Complex Networks
3. Mobile Networks Security
4. Web Application Security
5. AI for Cyber Security
6. Cyber Security for Adversarial AI
7. Cloud Computing Security
8. Usable Security and Privacy
9. Smart Devices and IoT Forensics
10. Cyber Operations
11. Human Factors in Cyber Security
12. Cyber Warfare
13. Mathematics for Cyber Security
14. Advanced Cryptanalysis
15. Blockchain and Web3.0
16. Quantum Cryptography
17. Embedded Systems Security
18. Cyber-Physical Systems Security
19. Financial Systems Security
20. Space Systems Cyber Security
21. Advanced Information Assurance
22. Advanced Operating System Security
23. Advanced Topics in Cryptography
24. Advanced Wireless Networks Security
25. Advanced Cyber Threat Intelligence
26. Advanced Ethical Hacking and Defence
27. Advanced Computer Forensics
28. Advanced Database Security

29. Advanced Malware Analysis
30. Advanced Vulnerability Assessment and Reverse Engineering

DEGREE AWARD REQUIREMENTS

The following minimum requirements are prescribed for the award of a MS Cyber Security:

- a) A minimum of 26 credit hours of course work including **09 credit hours for core courses, 15 credit hours for specialization-oriented electives and 2 credit hours for Quranic courses** as prescribed in this document must be completed.
- b) In addition, research thesis of 6 credit hours or course work of 3 courses of 9 credit hours must also be completed individually as partial fulfilment of the degree program.
- c) CGPA must not be below 2.50 / 4.00 at the time of completion of the degree program. The university may, however, set a higher standard in this regard.
- d) The minimum duration required to complete the degree is 3 regular semesters (1.5 Years) and maximum duration is 8 regular semesters (4 Years) . Summer semester is not considered as a regular semester.

Save exceptions: In case a student is unable to secure an MS within the prescribed timeframe and claims for extension in duration, the university may constitute appropriate authority and determine the causes of delay. In the event of force majeure (i.e., delay on account of circumstance beyond the control of student), the university may grant an extension in the period of award of MS degree in accordance with the duration limiting factor(s) and shall also take corrective measures in case the delay is caused by process or administrative reasons.

COURSE LEARNING OUTCOMES (CLOS):

(MANDATORY MAJOR COURSES BS CYBER SECURITY)

1. Programming Fundamentals

By the end of this course student will be able to:

- Understand the fundamental concepts of programming
- Apply basic programming concepts using a programming language
- Develop programs for data processing and file handling
- Use user-defined data types to solve real-world computing problem

2. Object Oriented Programming

By the end of this course student will be able to:

- Understand principles of the object-oriented paradigm
- Apply the object-oriented principles to solve real-world problems
- Identify the objects & their relationships to build an object-oriented solution
- Solve computing problems in an object-oriented programming language
- Implement a project to solve a real-world computing problem in an object-oriented programming language.

3. Introduction to Cyber Security

By the end of this course student will be able to:

- Explain fundamental concepts, principles, and terminology of cybersecurity, including emerging threats.
- Identify, classify, and analyze cyber attacks and vulnerabilities in modern networks and systems.
- Demonstrate awareness of ethical, legal, and professional issues in cybersecurity.
- Apply basic risk assessment techniques to real-world scenarios.

4. Data Structures and Algorithms

By the end of this course student will be able to:

- Apply data structures (arrays, lists, trees, graphs) in cybersecurity applications such as intrusion detection and threat analysis.
- Design algorithms for efficient data processing, cryptography, and log analysis.
- Analyze algorithm performance using time and space complexity for scalable solutions.
- Implement data structures to support security applications, including malware detection and traffic analysis.

5. Database Systems

By the end of this course student will be able to:

- Design and implement secure relational databases and queries for storing sensitive data.
- Apply SQL and NoSQL techniques for data retrieval and management.
- Implement access control, encryption, and integrity checks for database security.
- Analyze and mitigate database vulnerabilities in enterprise environments.

6. Digital Logic Design

By the end of this course student will be able to:

- Design and analyze combinational and sequential digital circuits with security considerations.
- Apply Boolean algebra and minimization techniques to optimize hardware security.
- Simulate and implement digital circuits using modern software and hardware tools.
- Understand hardware design principles relevant to embedded and IoT security.

7. Computer Networks

By the end of this course student will be able to:

- Explain networking concepts, protocols, and topologies with a focus on security.
- Configure and secure network devices, including firewalls and VPNs, against cyber threats.
- Analyze network traffic and logs to detect vulnerabilities and potential attacks.
- Apply encryption, authentication, and secure communication methods for modern networks.

8. Operating Systems

By the end of this course student will be able to:

- Explain OS architecture, process management, memory, and file systems in the context of security.
- Apply system hardening techniques to secure Windows and Linux environments.
- Implement access control policies, authentication, and permission models.
- Perform OS-level troubleshooting and incident analysis for secure computing environments.

9. Information Security Management

By the end of this course student will be able to:

- Explain principles, frameworks, and best practices of information security management.
- Develop organizational security policies, procedures, and risk mitigation strategies.
- Evaluate security risks and propose controls for enterprise and cloud environments.
- Assess compliance with international standards (ISO 27001, NIST, GDPR) and governance models.

10. Artificial Intelligence (Mandatory IDS-I)

By the end of this course student will be able to:

- Explain AI concepts and algorithms relevant to cybersecurity applications.
- Apply machine learning models to detect anomalies, malware, and network threats.
- Analyze datasets for predictive threat intelligence and automated decision-making.
- Implement AI-based security solutions to support defensive and offensive operations.

11. Secure Software Design and Development

By the end of this course student will be able to:

- Apply secure software development lifecycle (SDLC) principles in real-world applications.
- Identify and mitigate software vulnerabilities (e.g., SQLi, XSS, buffer overflows).
- Implement secure coding standards and perform automated security testing.
- Design software architectures with integrated security controls for cloud and web applications.

12. Computer Organization and Architecture

By the end of this course student will be able to:

- Explain computer architecture concepts and hardware-software interaction in secure computing.
- Analyze the impact of CPU, memory, and I/O design on system security.
- Implement low-level programs to understand hardware-level security implications.
- Apply knowledge of memory hierarchy and instruction execution to security-critical systems.

13. Digital Forensics and Incident Response

By the end of this course student will be able to:

- Explain digital forensics principles, techniques, and legal considerations.
- Collect, preserve, and analyze evidence from computers, networks, and IoT devices.
- Perform incident response using modern tools and frameworks (SOAR, forensic suites).
- Apply threat intelligence and OSINT techniques to investigate cyber incidents.

14. Cryptography

By the end of this course student will be able to:

- Explain principles, algorithms, and protocols of symmetric and asymmetric cryptography.
- Implement encryption, decryption, hashing, and digital signatures for data security.
- Analyze cryptographic systems for vulnerabilities and potential attacks.
- Apply cryptography in cloud, blockchain, and post-quantum computing contexts.

15. Vulnerability Assessment and Penetration Testing

By the end of this course student will be able to:

- Conduct vulnerability assessments of networks, applications, and devices using modern tools.
- Perform penetration testing following ethical hacking methodologies.
- Document findings and recommend remediation strategies.
- Apply automated tools and manual techniques for security testing in cloud and enterprise systems.

16. Malware Analysis and Reverse Engineering

By the end of this course student will be able to:

- Analyze malware behavior and lifecycle in modern computing environments.
- Perform static and dynamic analysis of malicious code using appropriate tools.
- Reverse engineer software to detect vulnerabilities and exploit techniques.
- Develop mitigation strategies to protect systems from advanced persistent threats (APTs).

17. Cloud and Application Security

By the end of this course student will be able to:

- Explain security architectures, shared responsibility models, and threat vectors in cloud-native and web-based applications.
- Apply secure configuration and hardening controls for cloud services, APIs, and application deployment pipelines.
- Analyze application logs, cloud telemetry, and security alerts to identify misconfigurations, vulnerabilities, and active attacks.
- Evaluate security weaknesses in applications and cloud environments using risk, impact, and exploitability criteria.
- Design mitigation and monitoring strategies that support both prevention and post-incident investigation.

(SPECIALIZATION COURSES- BS CYBER SECURITY)

1) Network and Infrastructure Security

Cloud Security

By the end of this course student will be able to:

- Explain fundamental cloud computing models (IaaS, PaaS, SaaS) and identify security risks and responsibilities in public, private, and hybrid cloud environments.
- Analyze common cloud security threats and evaluate appropriate security controls for protecting cloud-based infrastructure and services.
- Apply cloud security mechanisms such as identity and access management (IAM), encryption, network security groups, and monitoring tools to secure cloud resources.
- Design and implement a secure cloud architecture that follows shared responsibility principles, compliance requirements, and cloud security best practices.

Data Center Security

By the end of this course student will be able to:

- Explain data center architectures, components, and operations, and identify security risks across physical and logical layers.
- Analyze data center security threats and failures, and assess their impact on system availability, integrity, and confidentiality.
- Apply security controls such as access control, segmentation, monitoring, backup, and recovery to protect data center infrastructure.
- Design and implement a secure data center environment that supports resilience, compliance, and business continuity.

Mobile and Wireless Network Security

By the end of this course student will be able to:

- Explain the architecture and protocols of mobile and wireless networks, and identify related security threats and vulnerabilities.

- Analyze attacks on mobile and wireless networks and assess their impact on confidentiality, integrity, and availability.
- Apply security mechanisms and standards such as authentication, encryption, key management, secure routing, and access control to protect wireless communications and mobile devices.
- Design and implement a secure mobile or wireless network solution that follows established standards and risk-based security practices.

Network and System Administration

By the end of this course student will be able to:

- Explain the roles of network and system administrators, and describe the architecture and operation of servers, operating systems, and enterprise networks.
- Manage network services and system resources, including users, storage, addressing, and core services, following accepted practices.
- Apply security and reliability measures such as access control, patching, backup, monitoring, and hardening to maintain system stability and security.
- Troubleshoot network and system issues related to performance, availability, configuration, and security using appropriate tools.

Machine Learning for Network Security

By the end of this course student will be able to:

- Explain core machine learning concepts and algorithms, and describe their relevance to network security problems.
- Analyze network traffic and security datasets to identify patterns, anomalies, and attack behaviors using appropriate data processing techniques.
- Apply machine learning models to detect intrusions, malware, and anomalous network activity, and evaluate their performance using standard metrics.
- Design and implement an ML-based network security solution that considers data collection, model training, deployment, scalability, and false-positive control.

Virtualization Security

By the end of this course student will be able to:

- Explain the core concepts and architectures of SDN, NFV, and SDR, and identify their associated security challenges.
- Analyze security threats and attack surfaces in SDN, NFV, and SDR environments, and assess their potential impact on network operations.
- Apply security mechanisms and controls for SDN controllers, NFV infrastructure, and SDR platforms to protect network services and communications.
- Design and implement a secure and resilient integrated environment using SDN, NFV, and SDR while considering performance, scalability, and regulatory constraints.

Network Forensics

By the end of this course student will be able to:

- Explain network forensics concepts and processes, including evidence acquisition, preservation, and chain of custody.
- Analyze network traffic, logs, and flow data to identify security incidents, attack patterns, and malicious activities.
- Prepare clear and structured forensic reports that present findings and conclusions for technical, managerial, and legal audiences.

- Apply network forensic tools and techniques to collect, examine, and reconstruct network-based incidents.

Endpoint Security

By the end of this course student will be able to:

- Explain key concepts of endpoint security, including threats, attack surfaces, and security principles for endpoint devices and operating systems.
- Analyze common endpoint threats and vulnerabilities, and assess their impact on confidentiality, integrity, and availability.
- Apply endpoint security mechanisms such as access control, authentication, hardening, patching, and malware protection to secure endpoint devices.
- Evaluate endpoint security using risk assessment, vulnerability analysis, and incident response procedures, and recommend mitigation strategies.

Network Security Monitoring

By the end of this course student will be able to:

- Explain the objectives and architecture of network security monitoring, including sensors, logs, flows, and centralized systems.
- Analyze network traffic and security events to detect anomalies, intrusions, and policy violations, and assess their impact on network operations.
- Apply network security monitoring tools and techniques such as traffic capture, flow analysis, intrusion detection, and log correlation to identify malicious activity.
- Interpret and report monitoring outcomes through alerts, dashboards, and incident summaries to support incident response and operational decision-making.

Security Automation for Network Operations

By the end of this course student will be able to:

- Explain the role of security automation and orchestration in network operations, including key benefits, limitations, and operational use cases.
- Analyze network security workflows to identify processes suitable for automation, such as alert handling, configuration tasks, and incident response.
- Apply automation techniques using scripts, APIs, and security tools to support network security monitoring, response, and policy enforcement.
- Design and implement an automated security workflow that integrates monitoring, detection, and response while considering reliability, scalability, and security.

2) Cryptography

Information Privacy and Security

By the end of this course, students will be able to:

- Explain core concepts of information privacy and data protection.
- Apply privacy principles and controls in information systems.
- Analyze privacy risks in organizational data environments.
- Evaluate legal and technical measures for privacy protection.

Mathematical Foundations for Cryptography

By the end of this course, students will be able to:

- Explain mathematical concepts used in cryptography.
- Apply number-theoretic and algebraic techniques in cryptographic problems.
- Analyze mathematical structures underlying cryptographic schemes.
- Solve cryptography-related problems using formal reasoning.

Symmetric Key Cryptography

By the end of this course, students will be able to:

- Explain principles of symmetric encryption algorithms.
- Apply symmetric-key algorithms for secure communication.
- Analyze block cipher structures and security properties.
- Evaluate modes of operation against cryptographic attacks.

Asymmetric Key Cryptography

By the end of this course, students will be able to:

- Explain principles of public-key cryptography.
- Apply asymmetric algorithms for encryption and digital signatures.
- Analyze computational assumptions of public-key schemes.
- Compare asymmetric algorithms based on security and efficiency.

Cryptographic Hash Functions and Blockchain Applications

By the end of this course, students will be able to:

- Explain properties and applications of cryptographic hash functions.
- Apply hash functions and MACs for data integrity and authentication.
- Analyze common attacks on hash-based constructions.
- Evaluate the role of hash functions in blockchain systems.

Cryptographic Protocols and Key Management

By the end of this course, students will be able to:

- Explain cryptographic protocols and key management mechanisms.
- Apply secure key exchange and authentication protocols.
- Analyze vulnerabilities in cryptographic protocols.
- Design secure protocols for given cybersecurity scenarios.

Cryptanalysis and Attack Techniques

By the end of this course, students will be able to:

- Explain cryptanalytic models and attack methodologies.
- Apply basic cryptanalysis techniques to cryptographic systems.
- Analyze side-channel and implementation-based attacks.
- Evaluate cryptographic algorithms against known attack techniques.

Secure Systems and Applied Cryptography

By the end of this course, students will be able to:

- Explain the use of cryptography in secure systems.
- Apply cryptographic libraries and APIs in applications.
- Analyze common cryptographic implementation flaws.
- Develop secure software using cryptographic best practices.

Advanced Cryptography

By the end of this course, students will be able to:

- Explain advanced cryptographic concepts and constructions.
- Analyze modern cryptographic schemes.
- Evaluate cryptographic techniques under advanced threat models.
- Design cryptographic solutions for complex security problems.

Post-Quantum and Emerging Cryptography

By the end of this course, students will be able to:

- Explain the impact of quantum computing on cryptography.
- Apply post-quantum cryptographic algorithms.
- Analyze security and performance trade-offs of post-quantum schemes.
- Evaluate readiness for post-quantum cryptographic transition.

3) Digital Forensics and Incident Response

Computer Forensics

By the end of this course student will be able to:

- Explain forensic investigation models, evidence types, and standards applicable to modern operating systems.
- Perform forensically sound acquisition of storage media and volatile memory.
- Analyze file systems, registry entries, logs, and user artifacts to reconstruct system activity.
- Assess evidence integrity through hashing, timelines, and chain-of-custody verification.
- Prepare defensible forensic reports suitable for legal and organizational proceedings.

Network Forensics

By the end of this course student will be able to:

- Explain network protocols, traffic flows, and logging sources relevant to forensic investigations.
- Capture, filter, and reconstruct network sessions from live and recorded traffic.
- Analyze packet data and flow records to identify intrusions and lateral movement.
- Correlate network artifacts with host evidence to reconstruct attack timelines.

Smart Devices and IoT Forensics

By the end of this course student will be able to:

- Explain the unique challenges and legal considerations in acquiring evidence from smart and IoT devices.
- Apply appropriate tools and methods to extract data from smartphones (Android/iOS) and IoT devices.
- Analyze extracted mobile data (call logs, messages, location data, app artifacts) to support an investigation.
- Assess the integrity and reliability of data extracted from mobile devices using different acquisition methods.

Cloud Forensics and Log Analysis

By the end of this course student will be able to:

- Explain forensic implications of IaaS, PaaS, and SaaS cloud service models.
- Collect cloud artifacts including audit logs, access records, and snapshots.
- Analyze multi-source cloud logs to detect unauthorized access and data misuse.
- Evaluate jurisdictional, multi-tenancy, and data ownership challenges in cloud forensics.
- Produce structured cloud investigation reports aligned with incident response objectives.

Cyber Threat Intelligence

By the end of this course student will be able to:

- Explain the threat intelligence lifecycle and intelligence consumption models.
- Collect threat data from OSINT, technical feeds, and incident artifacts.

- Analyze threat actor TTPs using structured frameworks (e.g., ATT&CK).
- Produce actionable intelligence products tailored to operational and strategic use.

Open-Source Intelligence Techniques

By the end of this course student will be able to:

- Explain OSINT principles, scope, and related ethical and legal considerations.
- Analyze tools, methods, and frameworks for gathering and validating intelligence from diverse open sources.
- Evaluate OSINT applications across domains such as law enforcement, journalism, finance, and cybersecurity, noting risks and best practices.
- Apply OSINT techniques, including SOCMINT and dark web exploration, to conduct safe and effective investigations.

Incident Response Orchestration and Automation

By the end of this course student will be able to:

- Explain incident response phases, roles, and automation principles.
- Configure and execute automated response workflows using SOAR platforms.
- Analyze alerts and telemetry to trigger response playbooks.
- Evaluate response effectiveness using metrics such as containment time and impact reduction.
- Design and refine incident response playbooks based on lessons learned.

Multimedia Forensics

By the end of this course student will be able to:

- Explain digital image, audio, and video formats and manipulation techniques.
- Extract and examine multimedia metadata and content artifacts.
- Analyze media files to detect tampering, forgery, or deepfake indicators.
- Assess the evidentiary reliability of multimedia artifacts in investigations.

eDiscovery Forensics and Moot Courts

By the end of this course student will be able to:

- Explain legal, procedural, and technical foundations of eDiscovery.
- Conduct identification, preservation, and collection of electronically stored information (ESI).
- Analyze digital evidence within discovery, review, and production workflows.
- Evaluate admissibility and presentation of digital evidence under legal standards.
- Demonstrate expert witness skills through moot court simulations.

AI-Driven Digital Forensics

By the end of this course student will be able to:

- Explain AI and machine learning techniques applicable to digital forensic analysis.
- Apply AI-based methods for evidence classification, clustering, and anomaly detection.
- Analyze large-scale forensic datasets using automated and intelligent techniques.
- Evaluate accuracy, bias, explainability, and legal risks of AI-driven tools.
- Design AI-assisted forensic workflows to enhance investigative efficiency.

4) Embedded Systems Security

Introduction to Embedded System Security

By the end of this course student will be able to:

- Explain the architecture of embedded systems and identify security vulnerabilities across hardware, firmware, and communication interfaces.
- Analyze common threats and attack vectors targeting embedded.
- Apply fundamental security mechanisms such as secure boot, cryptography etc to protect embedded systems.
- Evaluate embedded system security using basic risk assessment, testing, and mitigation techniques considering resource constraints.

Internet of Things(IoT) Security

By the end of this course student will be able to:

- Describe IoT architectures, protocols, and ecosystems, and identify security and privacy all layers.
- Analyze common IoT threats and attack vectors, including device compromise, insecure communication, cloud vulnerabilities, and supply-chain risks.
- Apply appropriate security mechanisms such as authentication, encryption, key management, and secure firmware updates to protect IoT systems.
- Evaluate IoT deployments using risk assessment, and recommend mitigation strategies for real-world scenarios.

Mobile Device Security

By the end of this course student will be able to:

- Explain mobile device architectures, operating systems, and identify security threats and vulnerabilities affecting smartphones and tablets.
- Analyze common mobile attack vectors such as malicious applications, insecure wireless communications, phishing, and device loss or theft.
- Apply mobile security mechanisms including secure authentication, encryption, application sandboxing, and mobile device management (MDM) controls to protect mobile platforms.
- Evaluate mobile applications and devices using security testing, risk assessment, and privacy considerations.

Industrial IoT and Infrastructure Security

By the end of this course student will be able to:

- Explain Industrial IoT (IIoT) architectures, components, and communication protocols, and identify security challenges.
- Analyze cyber and cyber-physical threats targeting IIoT systems, including attacks on PLCs, SCADA/ICS networks, sensors, and industrial communication links.
- Apply appropriate security controls such as network segmentation, secure communication, access control, and monitoring to protect industrial infrastructure.
- Evaluate the security posture of IIoT deployments using risk assessment, standards, and incident response considerations, and propose mitigation strategies.

Smart and Autonomous Vehicle Security

By the end of this course student will be able to:

- Explain the architecture of smart and autonomous vehicles, identify security challenges of in-vehicle networks, sensors, and V2X communication.
- Analyze cyber, physical, and cyber-physical attack vectors targeting autonomous vehicle systems, such as CAN bus attacks, sensor spoofing, and AI model manipulation.
- Apply automotive security mechanisms including secure communication, authentication, and cryptography to protect vehicle systems.
- Evaluate the security and safety of smart and autonomous vehicles using risk assessment, standards, and testing techniques, and recommend mitigation strategies.

Avionics Security

By the end of this course student will be able to:

- Explain the architecture, communication systems, and security requirements of avionics and aerospace digital platforms.
- Analyze cyber threats, vulnerabilities, and risk scenarios affecting airborne and ground-based aviation systems.
- Apply security controls and assurance mechanisms to protect avionics networks, software, and data flows.
- Evaluate the effectiveness of aviation cybersecurity measures in alignment with international standards, safety, and regulatory frameworks.

Smart Devices and IoT Security

By the end of this course student will be able to:

- Explain the architecture, threat landscape, and attack surfaces of smart and connected devices.
- Apply acquisition and analysis techniques to collect, preserve, and interpret digital evidence from device hardware, firmware, and network interactions.
- Conduct structured penetration testing of smart devices using standard frameworks and reporting methodologies.
- Evaluate security findings and recommend mitigation strategies in compliance with legal, ethical, and professional standards.

Hardware Design Security

By the end of this course student will be able to:

- Describe hardware architectures, threat models, and common vulnerabilities in hardware designs.
- Analyze the impact of physical, side-channel, and hardware-based attacks on system security and reliability.
- Design secure hardware components using principles of trusted computing and defense-in-depth.
- Assess hardware security controls and recommend improvements aligned with industry standards and best practices.

Embedded Operating System Security

By the end of this course student will be able to:

- Explain the architecture and security requirements of embedded and real-time operating systems.
- Implement core security mechanisms such as secure boot, process isolation, and access control.
- Analyze vulnerabilities and threat scenarios affecting embedded OS platforms.
- Evaluate the effectiveness of embedded OS security controls using structured testing and validation approaches.

Wireless Networks Security

By the end of this course student will be able to:

- Describe wireless network architectures, protocols, and associated security threats.
- Configure authentication, encryption, and access control mechanisms for secure wireless communications.
- Analyze wireless attacks and vulnerabilities using monitoring and assessment tools.

- Design secure wireless network solutions aligned with industry standards and regulatory requirements.

5) Governance, Risk and Compliance

Cyber Governance

After completing this course, students will be able to:

- Explain core concepts, principles, and frameworks of cyber governance.
- Identify legal and regulatory requirements related to cybersecurity.
- Assess cyber risks and governance controls within organizations.
- Develop cyber governance policies aligned with organizational goals.

Cyber Security Risk Management

After completing this course, students will be able to:

- Explain concepts and processes of cybersecurity risk management.
- Identify and analyze cyber threats, vulnerabilities, and their potential impacts.
- Apply risk assessment and risk treatment techniques to cybersecurity scenarios.
- Recommend risk mitigation and monitoring strategies aligned with organizational objectives.

Cyber Warfare

After completing this course, students will be able to:

- Explain the concepts, actors, and doctrines of cyber warfare and cyber conflict.
- Analyze cyber warfare tactics, techniques, and operations used in modern conflicts.
- Assess the impact of cyber warfare on national security and critical infrastructure.
- Evaluate legal, ethical, and strategic considerations in cyber warfare operations.

Cyber Laws, Regulations & Ethics

After completing this course, students will be able to:

- Describe key national and international cyber laws.
- Explain legal and ethical issues related to cyberspace.
- Interpret privacy and data protection requirements.
- Analyze legal implications of cyber incidents.

Compliance Standards & Frameworks

After completing this course, students will be able to:

- Explain major cybersecurity standards and frameworks.
- Compare ISO, NIST, COBIT, and other compliance models.
- Perform cyber security compliance gap analysis.
- Map organizational controls to relevant standards.

Cyber Security Audit & Assurance

After completing this course, students will be able to:

- Explain principles of information security auditing with related ISO and other standards.
- Conduct cyber and information security control assessments.
- Collect, document, and evaluate appropriate audit evidence using cybersecurity audit techniques and tools to assess the effectiveness of security controls, compliance with standards, and assurance of information systems

- Prepare an audit report with findings and recommendations.

Financial Systems Security

After completing this course, students will be able to:

- Explain the architecture, operations, and security requirements of financial systems.
- Identify and analyze cyber threats, vulnerabilities, and fraud risks in financial environments.
- Apply security controls, standards, and regulatory requirements to protect financial data and transactions.
- Develop and evaluate business continuity and disaster recovery strategies for financial systems to ensure service resilience.

Privacy Engineering & Data Protection

After completing this course, students will be able to:

- Explain core principles of data privacy and protection.
- Apply privacy-by-design concepts in system development.
- Classify and handle sensitive data appropriately.
- Analyze organizational compliance with data protection laws.

Third-Party & Supply Chain Risk Management

After completing this course, students will be able to:

- Identify risks arising from third-party and vendor relationships.
- Perform basic vendor security risk assessments.
- Evaluate security clauses in contracts and SLAs.
- Propose mitigation strategies for supply chain risks.

Human Behaviours in Cybersecurity

After completing this course, students will be able to:

- Explain the role of human behavior in cybersecurity incidents and defenses.
- Identify social engineering techniques and user-related security weaknesses.
- Analyze psychological, cultural, and organizational factors affecting cybersecurity behavior.
- Propose awareness, training, and behavioral interventions to reduce human-related cyber risks.

(MANDATORY COURSES MS CYBER SECURITY)

1) Data and Software Security

After completing this course, students will be able to:

- Understand threats to data at rest, in transit, and in processing, and the technologies used to mitigate them.
- Analyze security scenarios to identify data-related threats and propose solutions integrating people, processes, and technology.
- Understand software security threats across the SDLC and describe frameworks and techniques for secure software development.
- Conduct a literature survey in data and software security and synthesize current research in the form of a survey article.

2) Advanced Network Security

After completing this course, students will be able to:

- Understand advanced network threats, vulnerabilities, and defense mechanisms across modern network architectures.
- Analyze complex network security scenarios and propose integrated security solutions.
- Evaluate advanced network security protocols, architectures, and intrusion detection and response mechanisms.
- Conduct a literature survey on an advanced network security topic and synthesize state-of-the-art research.

3) Research Methodology

After completing this course, students will be able to:

- Understand foundational and advanced research paradigms, methodologies, and ethical considerations relevant to cybersecurity and computing research.
- Formulate well-defined research problems, objectives, and research questions, and select appropriate qualitative, quantitative, or mixed research methods and research designs.
- Critically analyze and evaluate scholarly literature to identify research gaps, limitations, trends, and opportunities within cybersecurity domains.
- Design and conduct a systematic literature review and synthesize findings into a structured research proposal or a research paper following standard publication practices.

FIELD EXPERIENCE / INTERNSHIP GUIDELINES – BS CYBER SECURITY PROGRAM

1. Field Experience / Internship Requirement

- Field Experience / Internship is **mandatory for all students** enrolled in the BS Cyber Security program.
- The internship carries **3 credit hours**, equivalent to a **minimum of 150 working hours**.
- Students may undertake the internship **after successful completion of the first year** of the degree program, subject to university regulations.

2. Accepted Modes of Internship

Internships can be completed through any of the following approved modes:

- **On-site (Physical):** Working at the premises of a recognized organization, industry, or institution.
- **Remote / Online:** Working virtually for an approved organization under proper supervision and reporting mechanisms.
- **Industry- or University-Assigned Projects/Workshops:** Completion of relevant, approved projects, training programs, or workshops assigned by the university in collaboration with industry or internal departments.

3. Evaluation and Assessment

- The university shall establish a **formal evaluation and assessment mechanism** to ensure the quality and relevance of the internship experience.
- Assessment may include, but is not limited to:
 - Internship report and/or logbook
 - Evaluation and feedback from the industry supervisor or mentor
 - Oral presentation and/or viva voce
 - Assessment of project outcomes or assigned deliverables

Framework for Integration of International Professional Certifications into BS Cyber Security Program

1. Title

Integration of International Professional Certifications into BS Cyber Security Program.

2. Objectives

- To **recognize selected international professional certifications** as equivalent to **one elective course (3 credit hours)** within the BS Cyber Security programs.
- To ensure that the recognition of certifications **maintains academic rigor** and **does not compromise NCEAC accreditation standards**.
- To **enhance graduate employability** by complementing formal academic education with verifiable industry recognized credentials.
- To ensure the list of recommended certifications as proposed by the *National Curriculum Review Committee (NCRC) for Cyber Security* or Pakistan Software Export Board (PSEB), is **approved by the relevant academic statutory bodies** of the university (Board of Studies, Board of Faculty and Academic Council) with the provision for universities to recommend additional certifications.
- To facilitate **equitable access** by enabling government and other approved funding mechanisms to support the cost of certifications for university students and affiliated colleges, subject to availability and policy guidelines.

3. Scope and Applicability

- Applicable to **all HEC-recognized universities/Degree Awarding Institutions (DAIs)** in Pakistan offering NCEAC-accredited BS Cyber Security programs.
- Effective from **Fall 2026 intake**.

4. Guiding Principles

- **Credit Recognition:** A maximum of 3 credit hours may be awarded per student upon successful completion of an approved certification, involving at least 48 hours of structured learning and engagement.
- **Category Restriction:** Certification-based credit shall be allowed only in lieu of one elective course and shall not replace any core or compulsory course.
- **Validation Requirement:** Prior to credit award, students must successfully pass a university-administered assessment, which may include a test, practical evaluation, or viva voce, to validate acquired competencies.
- **Verification Requirement:** All certifications must be verifiable with the issuing authority through official validation mechanisms.
- **Standardization Across Universities:** All participating universities shall implement this framework uniformly, in accordance with the guidelines approved by the National Curriculum Review Committee (NCRC).

5. Review and Continuous Improvement

The Higher Education Commission (HEC) shall review and update this framework every three (3) years, or earlier if required, to ensure continued relevance with evolving industry needs, academic standards, and accreditation requirements.

6. Recommended Cyber Security Certifications (Initial List – 2026)

1. (ISC)² CC - Certified in Cybersecurity
2. Google Cybersecurity Certificate -Coursera
3. Google Cloud Skills Boost -AI for Cybersecurity Quest
4. CompTIA Security+
5. CompTIA PenTest+
6. Cisco -Introduction to Cybersecurity
7. Cisco CCNA(200-301)
8. Cisco CyberOps Associate
9. Linux Essentials/ Linux Professional Institute Certification (LPIC-1)
10. Fortinet Certified Fundamentals- Network Security Expert 1 (NSE-1)
11. Fortinet Network Security Expert 2 (NSE-2)
12. TryHackMe –for hands-on practice
13. HackTheBox Academy
14. HackTheBox-CPTS – Certified Penetration Testing Specialist
15. IBM- Penetration Testing, Incident Response and Forensics
16. IBM Cybersecurity Analyst Professional Certificate
17. IBM QRadar SIEM Foundation
18. EC-Council- CEH – Certified Ethical Hacker
19. TCM Security -PNPT – Practical Network Penetration Tester
20. TCM Security -PJPT-Practical Junior PenTester
21. INE Security- eJPT – eLearnSecurity Junior Penetration Tester (eJPT)
22. INE Security- eWPTX
23. Coursera- AI for Cybersecurity Specialization- Johns Hopkins University
24. INFOSEC- Computer Forensics Specialization
25. Microsoft Security Fundamentals (SC-900)
26. Microspft Azure AI Fundamentals (AI-900)
27. Cybrary- Incident Response & Cyber Investigations
28. Qualys- Qualys Certified Courses
29. Splunk Core Certified User
30. AWS Certified Security – Specialty
31. PECB- ISO IEC 27001 Foundation
32. ISC²- Systems Security Certified Practitioner